Giseop KIM

Assistant Professor

Department of Robotics and Mechatronics Engineering, DGIST, South Korea
Joint appointments: Department of Artificial Intelligence; Mechanical Engineering Track
Leader/Director, Autonomy and Perceptual Robotics Lab (APRL)

gsk@dgist.ac.kr
Personal Website
Lab Website
Google Scholar
Lab YouTube

Research Group

Research mission. APRL advances spatial intelligence and robot world models to enable autonomous robots to navigate, reason, and learn from experience in the real world.

Research directions.

- **Spatial intelligence for navigation:** SLAM 2.0 for the Robot Web era; neural map representations; human-robot interactive visual-language navigation; long-term and multi-robot autonomy.
- **Embodied reasoning and world models:** Reasoning for robots; generative AI for mobile robot navigation; memory-augmented spatial intelligence; robot world models.

Research profile. Robotics researcher working at the intersection of simultaneous localization and mapping (SLAM), spatial and physical AI, 3D perception, sensor fusion, long-term autonomy, and visual-language robot navigation. Current research investigates memory-augmented spatial intelligence, shared robot experience, embodied reasoning, and robot world models for robust real-world navigation.

Lab composition. Autonomy and Perceptual Robotics Lab (APRL), DGIST. Founded December 2024. Current full-time group: one postdoctoral researcher, three doctoral researchers (including two integrated M.S./Ph.D. students), and four M.S. students, with undergraduate interns and external collaborators across robotics, perception, and AI.

Academic Appointments and Professional Experience

Dec. 2024–present	**Assistant Professor**, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu, Republic of Korea. Principal Investigator, APRL.
Dec. 2021–Dec. 2024	**Research Scientist**, NAVER LABS, Seongnam, Republic of Korea. Research in autonomous driving, 3D vision, mapping, and localization.
Mar. 2017–Aug. 2021	**Graduate Researcher**, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea. Intelligent Robotic Autonomy and Perception Lab.

Education

2022	**Ph.D., Civil and Environmental Engineering**, KAIST. Advisors: Prof. Ayoung Kim and Prof. Youngchul Kim.
2019	**M.S., Civil and Environmental Engineering**, KAIST.
2017	**B.S., Civil and Environmental Engineering**, KAIST.

Academic Leadership and Service

2026–present	**Associate Editor (AE)**, IEEE Robotics and Automation Letters — Localization and Mapping (RA-L).
2027	**Area Chair (AC)**, Robotics: Science and Systems (RSS 2027).
2027	**Workshops & Tutorials Committee Co-Chair**, IEEE International Conference on Robotics and Automation (ICRA 2027).
2026	**Workshop Organizer**, Second Workshop on Human-aware Embodied AI (IROS 2026).
2026	**Workshop Organizer**, Long-Term Perception for Human-Centric Autonomy (LTP): Towards Undo for the Physical World (IROS 2026).
2026	**Organizing Committee Member**, The 41st Annual Conference of the Institute of Control, Robotics, and Systems (ICROS 2026).
2026	**Tutorial Organizer and Speaker**, KRoC 1st Spatial AI Tutorial (KRoC 2026).

Conference Reviewer

2024, 2026, 2027	IEEE International Conference on Robotics and Automation (ICRA).
2023, 2024, 2025, 2026	IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
2024, 2025, 2026	Robotics: Science and Systems (RSS).
2025, 2026	IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
2026	European Conference on Computer Vision (ECCV).
2026	IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).
2026	Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track.
2027	AAAI Conference on Artificial Intelligence (AAAI).
2026	IEEE/SICE International Symposium on System Integration (SII).
2026	International Conference on Control, Automation and Systems (ICCAS).
2023, 2025	International Conference on Ubiquitous Robots (UR).
2026	International Conference on Space Robotics (iSpaRo).

Journal Reviewer

2023, 2024, 2026	IEEE Transactions on Robotics (T-RO).
2023, 2024, 2025, 2026	IEEE Robotics and Automation Letters (RA-L).
2026	The International Journal of Robotics Research (IJRR).
2025	IEEE Transactions on Aerospace and Electronic Systems.
2024	Advanced Robotics.

Honors and Awards

2026	**World's Top 2% Scientists**, single-year citation impact (2025), Stanford/Elsevier; Industrial Engineering & Automation, 1,341 of 131,085 (top 1.02%, excluding self-citations).
Sep. 2026	**Best Paper Award**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), for "LT-Mem: Volatility-Aware Spatio-Temporal Memory for Lifelong Scene Understanding." Recipients: Yumin Lee, Hyoseok Ju, and Giseop KIM.
Jul. 2026	**Outstanding Young Researcher Award**, 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026).
Feb. 2026	**Early-career Researcher Award**, Korea Robotics Society Annual Conference (KRoC 2026).
Oct. 2025	**Best Poster Award**, Human-aware Embodied AI Workshop at the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). Recipients: Jiseon Kim, Daeun Song, Bokeon Suh, Hyoseok Ju, Yumin Lee, Xuesu Xiao, and Giseop KIM.

Publications

Publication notation

►	First/co-first or corresponding-author paper by Giseop KIM.
*	First/co-first author.
†	Corresponding author.
Giseop KIM	The CV author's name is shown in bold.
"to appear"	Accepted paper awaiting publication.

Peer-Reviewed Journal Articles

J7. S. Lee*, H. Yu, **Giseop KIM**, and S. Choi†, "LAMP: Implicit Language Map for Robot Navigation," *IEEE Robotics and Automation Letters (RA-L)*, vol. 10, no. 12, pp. 12365–12372, 2025.

J6. M. Kim*, **Giseop KIM**, and S. Choi†, "Addressing Diverging Training Costs Using BEVRestore for High-Resolution Bird's Eye View Map Construction," *IEEE Robotics and Automation Letters (RA-L)*, 2024.

J5. H. Kim*, J. Choi, T. Sim, **Giseop KIM**, and Y. Cho†, "Narrowing Your FOV with SOLiD: Spatially Organized and Lightweight Global Descriptor for FOV-Constrained LiDAR Place Recognition," *IEEE Robotics and Automation Letters (RA-L)*, 2024.

J4. M. Jung*, W. Yang, D. Lee, H. Gil, **Giseop KIM**, and A. Kim†, "HeLiPR: Heterogeneous LiDAR Dataset for Inter-LiDAR Place Recognition under Spatiotemporal Variations," *The International Journal of Robotics Research (IJRR)*, vol. 43, no. 12, pp. 1867–1883, 2024.

J3. ▶ **Giseop KIM***, S. Choi, and A. Kim†, "Scan Context++: Structural Place Recognition Robust to Rotation and Lateral Variations in Urban Environments," *IEEE Transactions on Robotics (T-RO)*, 2022.

J2. Y. Cho*, **Giseop KIM**, S. Lee, and J.-H. Ryu†, "OpenStreetMap-based LiDAR Global Localization in Urban Environment without a Prior LiDAR Map," *IEEE Robotics and Automation Letters (RA-L)*, 2022.

J1. ▶ **Giseop KIM***, B. Park, and A. Kim†, "1-Day Learning, 1-Year Localization: Long-term LiDAR Localization using Scan Context Image," *IEEE Robotics and Automation Letters (RA-L)*, 2019.

Peer-Reviewed International Conference Papers

C19. ▶ H. Kim*, B. Kim, and **Giseop KIM**†, "MarsLab: A Martian Rover Simulator for Planetary Rover Autonomous Navigation," *International Conference on Space Robotics (iSpaRo)*, 2026, accepted, to appear.

C18. ▶ H. Kim* and **Giseop KIM**†, "Simulation for Planetary Robotic Perception and Autonomy: A Concise Survey of Recent Capabilities and Gaps," *International Conference on Space Robotics (iSpaRo)*, 2026, accepted, to appear.

C17. ▶ Y. Lee*, H. Ju, and **Giseop KIM**†, "LT-Mem: Volatility-Aware Spatio-Temporal Memory for Lifelong Scene Understanding," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Best Paper Award**.

C16. ▶ B. Nayak*, H. Ju*, and **Giseop KIM**†, "Mag4D-SLAM Dataset: A Repeated-Traversal Multi-Modal 4D Geomagnetic Dataset for Localization and Mapping," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

C15. S. Moon*, J. Baek, Y. Jeong, D. Chae, **Giseop KIM**, J. Lee, J. Kim†, and S. Choi†, "Streaming Dense Voxel Representations for 3D Occupancy Prediction," *European Conference on Computer Vision (ECCV)*, 2026.

C14. ▶ B. Suh*, H. Ju, and **Giseop KIM**†, "What Matters for Real-World Long-Horizon Robot Navigation?: An Experimental Study of Implicit Goals and Sparse Memory," *23rd World Congress of the International Federation of Automatic Control (IFAC)*, 2026.

C13. J. Lee*, J. Kim*, W. Shin, **Giseop KIM**, and H. Oh†, "Learning Point Cloud Geometry as a Statistical Manifold: Theory and Practice," *Robotics: Science and Systems (RSS)*, 2026.

C12. ▶ H. Ju*, B. Suh, and **Giseop KIM**†, "Have We Mastered Scale in Deep Monocular Visual SLAM? The ScaleMaster Dataset and Benchmark," *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

C11. M. Kim*, D. Lee, J. Yu, J. Hur, **Giseop KIM**, and J. Kim†, "Inlier-Centric Post-Training Quantization for Object Detection Models," *International Conference on Learning Representations (ICLR)*, 2026.

C10. J. Kim*, S. Jeong*, **Giseop KIM**, M.-H. Jeon, E. Jun, and A. Kim†, "2D Gaussian Splatting-based Sparse-view Transparent Object Depth Reconstruction via Physics Simulation for Scene Update," *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.

C9. H. Gil*, D. Lee*, **Giseop KIM**, and A. Kim†, "Ephemerality Meets LiDAR-based Lifelong Mapping," *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

C8. M. Kim*, **Giseop KIM**, K.-H. Jin, and S. Choi†, "BroadBEV: Collaborative LiDAR-Camera Fusion for Broad-Sighted Bird's Eye View Map Construction," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

C7. H. Lim*, K. Han, G. Shin, **Giseop KIM**, S. Hong, and H. Myung†, "ORORA: Outlier-Robust Radar Odometry," *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

C6. S. Yun*, M. Jung, J. Kim, S. Jung, Y. Cho, M.-H. Jeon, **Giseop KIM**, and A. Kim†, "STheReO: Stereo Thermal Dataset for Research in Odometry and Mapping," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

C5. ▶ **Giseop KIM*** and A. Kim†, "LT-mapper: A Modular Framework for LiDAR-based Lifelong Mapping," *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

C4. ▶ **Giseop KIM*** and A. Kim†, "Remove, then Revert: Static Point Cloud Map Construction using Multiresolution Range Images," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

C3. ▶ **Giseop KIM***, Y. S. Park, Y. Cho, J. Jeong, and A. Kim†, "MulRan: Multimodal Range Dataset for Urban Place Recognition," *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

C2. Y. Cho*, **Giseop KIM**, and A. Kim†, "Unsupervised Geometry-aware Deep LiDAR Odometry," *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

C1. ▶ **Giseop KIM*** and A. Kim†, "Scan Context: Egocentric Spatial Descriptor for Place Recognition within 3D Point Cloud Map," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.

Book Chapters

B1. J. Behley, M. Fallon, S. Zhao, **Giseop KIM**, J. Zhang, F. Zhang, and A. Kim, "LiDAR SLAM," Chapter 8 in *SLAM Handbook: From Localization and Mapping to Spatial Intelligence*, 2026.

Domestic Journal Articles

DJ1. ▶ B. Suh[*], J. Kim, J. Son, M. Park, and **Giseop KIM**[†], "Knowledge Fusion Strategies for Object Goal Navigation in the VLM Era: A Survey," *Transactions of the Korean Society of Automotive Engineers (Transactions of KSAE)*, vol. 34, no. 1, pp. 119–131, 2026.

Domestic Conference Papers

DC3. ▶ A. Lee[*] and **Giseop KIM**[†], "Floorplan-based Point-to-Language Grounded Navigation," Institute of Control, Robotics and Systems Annual Conference (ICROS), 2026.

DC2. ▶ A. Lee[*], H. Yoon[*], S. Park[*], M. Lee[*], and **Giseop KIM**[†], "Rendezvous Briefing: VLM-based Semantic-Aware Multi-Robot Collaborative Exploration," Institute of Control, Robotics and Systems Annual Conference (ICROS), 2026.

DC1. ▶ **Giseop KIM**[*†], M. Park, and J. Son, "Change-Robust LiDAR Localization in Urban Environments," Institute of Electronics and Information Engineers Conference, 2025.

Dissertations and Theses

T2. **Giseop KIM**, "LiDAR-based Lifelong Robotic Mapping in Changing Environments," Ph.D. dissertation, Korea Advanced Institute of Science and Technology (KAIST), 2022.

T1. **Giseop KIM**, "Isovist-induced Robust LiDAR Localization," M.S. thesis, Korea Advanced Institute of Science and Technology (KAIST), 2019.

Graduate Student Advising

DGIST	**Bokeon Suh**	Integrated M.S./Ph.D.	Fall 2025–present
DGIST	**Jiseon Kim**	M.S.	Fall 2025–present
DGIST	**Yumin Lee**	M.S.	Fall 2025–present
DGIST	**Hyoseok Ju**	Integrated M.S./Ph.D.	Fall 2025–present
DGIST	**Doyeon Kim**	Ph.D.	Spring 2026–present
DGIST	**Beomsu Kim**	M.S.	Spring 2026–present
DGIST	**Hoyun Kim**	M.S.	Spring 2026–present

Teaching

DGIST	2026 Fall	RT616	Vision/Image Processing
DGIST	2026 Fall	TM563	AI-based Autonomous Robot Systems
DGIST	2026 Spring	RT604	Advanced Mobile System
DGIST	2026 Spring	MECH301	Robots for Human
DGIST	2025 Fall	MECH307	Introduction to Artificial Intelligence
DGIST	2025 Fall	AT603	Introduction to Mobility Engineering
DGIST	2025 Spring	BE203	Creative Mechanical Design

Funded Projects

PI: Principal Investigator; Co-PI: Co-Principal Investigator; Participant: Participating Researcher.

Ongoing Projects

Project / funding details	Role	Period
Robot Industry Technology Development Program **Sponsor:** Ministry of Trade, Industry and Energy **Program:** 2026 Robot Industry Technology Development Program (Core Robot Industry Technology Development) **Title:** Spatial-RFM: Development of a VLM-based Spatial Intelligence Robot Foundation Model for Humanoid and Quadruped Robot Navigation in Unstructured Environments	Co-PI	Jul. 2026–Dec. 2029

Project / funding details	Role	Period
AIMS: Memory-Augmented Spatial Intelligence **Sponsor:** Ministry of Science and ICT **Program:** NRF Early-career Research Program (Type B), Basic Research Program 2026 **Title:** Memory-Augmented Spatial Intelligence for Heterogeneous Autonomous Mobility Systems Understanding and Interacting with Implicit Human Instructions	PI	Mar. 2026– Feb. 2030
Industry Project 2026-1 (Confidential) **Sponsor:** Industry-funded **Program:** Industry-academic technical service **Title:** Mapping and Localization in Factory Environments	PI	Apr. 2026– Mar. 2027
Basic Research Laboratory, Exploratory Type **Sponsor:** Ministry of Science and ICT **Program:** Basic Research Program, Basic Research Laboratory (Exploratory Type) 2026 **Topic:** Fundamental Study on Morphogenetic Cooperative Mobility Intelligence based on Physical Agentic AI	Co-PI	Jul. 2026– Jun. 2029
Glocal Lab **Sponsor:** Ministry of Education **Program:** Glocal Lab Program, Academic Research Support Program for Science and Engineering 2025 **Center:** TransHuman Robotics Research Center, Robotics and Mechatronics Research Institute of DGIST	Participant	Sep. 2025– Aug. 2028 (+6 years)
InnoCORE **Sponsor:** Ministry of Science and ICT **Title:** DGIST Team on Bio-Embodied Physical AI	Participant	Jul. 2025– Dec. 2027 (+3 years)
AI Star Fellowship **Sponsor:** Ministry of Science and ICT **Title:** Human-centered Symbiotic Embodied AI Systems	Participant	2025– 2030
DGIST Start-up Research **Sponsor:** DGIST **Title:** Self-evolving Robot-Web Navigation Intelligence	PI	Jan. 2025– Dec. 2028

Completed Projects

Project / funding details	Role	Period
P2P Autonomous Driving Platform **Sponsor:** Ministry of Trade, Industry and Energy **Program:** Autonomous Driving Technology Development Innovation Program, Korea Evaluation Institute of Industrial Technology **Title:** Development of a Level-4 Passenger-Van Autonomous Driving Platform for Point-to-Point Mobility in Designated Areas	Participant	Sep. 2025– Mar. 2026
N-HRHR **Sponsor:** DGIST **Title:** Exploring Human-Robot Collaborative Coexistence in the Era of Large-Scale Robots	Participant	Jul.– Dec. 2025

Invited Talks and Tutorials

Conference

Dec. 4, 2026 International — "Toward Efficient, Scalable, and Human-Like Robot Navigation Intelligence," invited talk, 14th International Conference on Robot Intelligence Technology and Applications (RiTA 2026), Daejeon, South Korea.

Sep. 27, 2026 International
"From Maps to Memory: Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation," Workshop on Bridging Navigation, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026).

Jul. 3, 2026 Domestic
"Visual Language Multi-robot SLAM," Early-career Researcher Award Session, 41st Annual Conference of the Institute of Control, Robotics and Systems (ICROS 2026), Daegu.

Feb. 6, 2026 Domestic
"IMU Basics and Inertial Aided Navigation," Spatial AI Tutorial Organizer and Speaker, Korea Robotics Society Annual Conference (KRoC 2026), Pyeongchang.

Feb. 5, 2026 Domestic
"Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation," Early-career Researcher Award Session, Korea Robotics Society Annual Conference (KRoC 2026), Pyeongchang.

Feb. 5, 2026 Domestic
"CoT, Scene Graphs, and Recent Paradigms for Mobile Robot Navigation: A Short Survey and Real-World Lessons," Special Session on AI-based Autonomous Systems, Korea Robotics Society Annual Conference (KRoC 2026), Pyeongchang.

Jan. 19, 2026 International
"An Invitation to 3D Vision, Spatial AI, and Physical AI," Invited Tutorial, International Conference on Electronics, Information, and Communication (ICEIC 2026), Macau.

Oct. 30, 2025 Domestic
"Physical AI for Smart Factory: Humanoid, Sensing, and Spatial Intelligence," Fall Conference of the Korean Fiber Society, Busan.

Aug. 15, 2025 Domestic
"Generative AI for Mobile Robot Navigation," AI Robotics KR Conference: Human and AI Robotics Together, Seoul.

Jul. 18, 2025 Domestic
"Generative AI for Mobile Robot Navigation," Joint Summer School on Image Understanding and Image Processing, Institute of Electronics and Information Engineers, Seoul.

University

Jul. 24, 2026 Domestic
"Bridging the Perception–Reasoning Gap for Long-Term Robot Navigation (Feat. the Research Journey)," Invited Talk, Korea Advanced Institute of Science and Technology (KAIST).

May 28, 2026 Domestic
"3D Perception, Language, and AI for Human-Robot Interactive Navigation," Ewha Womans University.

Apr. 2, 2026 Domestic
"Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation," Hanyang University ERICA.

Dec. 10, 2025 Domestic
"SLAM, Spatial AI, and Human-Robot Interactive Navigation," Department of Intelligent Robotics, Sungkyunkwan University, Suwon.

Dec. 9, 2025 Domestic
"Robotics and Spatial AI for Construction," Department of Civil and Environmental Engineering, KAIST, Daejeon.

Research Institute

Sep. 4, 2026 Domestic
"Lidar Inertial SLAM for Spaceship Docking," Invited Seminar, Korea Aerospace Research Institute (KARI).

Aug. 19, 2026 Domestic
"From Maps to Memory: Bridging Perception and Reasoning for Long-Term Autonomous Navigation," Invited Seminar, Korea Institute for Robot Industry Advancement (KIRIA), Seoul.

Jul. 30, 2026 Domestic
"Geometric 3D Vision Made Easy? – The DUSt3R Family and Its Evolution," Invited Seminar, Electronics and Telecommunications Research Institute (ETRI), Daegu-Gyeongbuk Regional Research Division.

Mar. 24, 2026 Domestic
"Memory-Augmented Spatial Intelligence for Autonomous Robot Navigation," ETRI Daegu-Gyeongbuk Regional Research Division.

Dec. 8, 2025 Domestic
"SLAM, Spatial AI, and Human-Robot Interactive Navigation," Department of Industrial Machinery DX, Korea Institute of Machinery and Materials, Daejeon.

May 28, 2025 Domestic
"Spatial AI, from the 2000s to 2025: The Evolution of Spatial Intelligence and the Future of Robotics," ETRI Daegu-Gyeongbuk Regional Research Division Mid- to Long-term Research Roadmap Meeting, Daegu.

Industry

Jun. 2, 2026 International
"What, When, and Why Visual-Language Navigation?," Private Meetup at the IEEE International Conference on Robotics and Automation (ICRA 2026), URobotics.

Jun. 13, 2025 Domestic
"Lessons from Translating Research into Products and Services," SLAM and Perception Seminar, Hyundai Motor Company Uiwang Research Institute.

Public Sector

Feb. 2, 2026 Domestic
"Daegu's Innovation, Moving Towards a Robot and Future Mobility City," Dalseo-gu Office Monthly Meeting Special Lecture, Daegu.